

VENKATA YASHWANT KUMAR BHYRI

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EDUCATION:

The Chinese University of Hong Kong, Shenzhen (*B.Eng. in Computer Science and Engineering*)

2023-27

- **Full Admission Excellence Scholarship** – School of Data science, CUHK-SZ
- **2024 Guangdong Government Outstanding International Student Scholarship**
- **2025 Leading Academic Peer Advisor @** School of Data Science, CUHK-SZ

TECHNICAL SKILLS:

Top Skills: Python, C++, SQL Hybrid, RISC-V, Git, Docker, Google GCP, AWS Deployment, Linux, Deployment Testing

AI Skills: OpenClaw, Google ADK, Agent System design, Prompt-Engineering, Fine-tuning, RL, Model Inference, TinyML

EXPERIENCE:

AI Agent Development Engineer [Intern] : AIGC Algorithms - *Wondershare Filmora* @ Shenzhen Jan 2026 – Present

- **Architected** and prototyped an end-to-end **Agent** based **AIGC** video generation and editing pipeline on **Google ADK** by implementing a unified seed-based generation workflow. Leveraged **seed diffusion** for regeneration with targeted **latent-space steering** to deliver real-time **AI editing** functionality with **character identity**, **scene geometry**, and **narrative coherence** across edits. Reduced **API token** overhead by ~**40%** and per-generation **cost** by ~**35%** versus full **Veo-3** resampling (~30–50% savings at the \$0.40/sec baseline).
- Engineered a **ML - feature-map control system** that translates orthogonal control axes (exposure, lighting tone, motion energy, transition speed, color palette, etc.) into **pixel-level semantic generation instructions** for **Google Veo 3 seed-regeneration**, enabling targeted **edits to specific video frame segments** without disturbing the video manifold. (*Embedding manipulation without full model re-sampling*)
- Built a semantic UI-to-latent translation interface that maps intuitive editing controls (sliders, palettes, tone adjustments) to **diffusion** conditioning vectors, enabling non-technical users to use AI editing features to **perform controlled semantic AI-video regeneration** for **multimodal alignment** across video, music, and transitions while **minimizing token-heavy LLM orchestration**.

AI Engineer Intern : AI Model Developer - Optek Microelectronics @ Shenzhen, China

2025 July - Sept

- Engineered a **full-stack TinyML Audio Classification Pipeline**, by integrating MediaPipe Audio for real-time feature extraction (log-Mel spectrograms, MFCCs), **TensorFlow Lite-Micro INT8** for quantized inference, and Edge Impulse for SDK deployment, achieving top notch-industry standards of **95%+ classification accuracy** on AudioSet, UrbanSound8K.
- **Optimized** and delivered a custom classifier for a **700 MHz DSP + 16 MB NPU** in the proprietary **XNCC** development environment, accomplishing **<10 ms latency**, **200 KB** runtime footprint, and **4×** model **compression** via INT8 quantization and pruning for scalable, low-power, **low latency** AI application.

Research Assistant : *HKU- COLUMBIA- ALIBABA- CUHKSZ* @ *BIRD Vision*

2025 June - Sept

- Reconstructed an advanced **multi-modal** benchmark framework that pioneered **BIRD-SQL** dataset (12,751+ question-SQL pairs, 95+ large databases) by integrating **vision + NLP** annotation for advance database queries and designing evaluation protocols to identify **model failure points**.
- Designed **relational DB** schemas, and created complex **hybrid SQL** queries to introduce reasoning challenges for vision-language models. Primarily involved in a robust **benchmark** construction, using schema-aware **prompt engineering**, **OCR** validation, annotating complex multimodal inputs.

Research Assistant – AI Model Benchmark Optimization @ CUHKSZ Natural Language Processing Group

2024

- Assisted to **automate** critical data-driven **Python pipelines** (Pandas, NumPy, Scikit-learn) for **hallucination detection**, **data grounding** index, reducing factual deviation by **31%** in all AI-generated benchmarked outputs. Engineered systematic benchmarking for **200+** LLM response pairs, boosting truth-layer **accuracy** by **25%**.

Research Assistant - Algorithmic Performance Optimization

2024

- Worked on a **novel script** framework to quantitatively **optimize algorithmic** - Augmented data retrieval (**RAG**) performance of an **ALLaVA** vision-language model. Improved Algorithmic data retrieval by **46%**, and optimized workflow operations of a **1.3 M**-sampled: labeled evaluation database.